

FINAL YEAR PROJECT

SYNOPSIS

on

*Reversible Data Hiding in Encrypted Medical Images Based on
Huffman Tree Coding and Count-Encryption*



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1. Introduction

Digital medical imaging supports diagnosis, treatment planning, follow-up, and remote collaboration. At the same time, the sharing and storage of such images introduces risks of unauthorised disclosure, inappropriate access, accidental mismatch between an image and its associated clinical note, and undetected alteration. Reversible data hiding is particularly relevant to medical images because it aims to embed auxiliary information while enabling exact recovery of the original image after extraction .

The baseline research paper, Reversible Data Hiding in Encrypted Medical Images Based on Huffman Tree Coding and Count-Encryption, uses prediction errors, canonical/adjusted Huffman-tree coding, and a comparison of Huffman Coding (HC) and Leaf-to-Leaf Coding (LLC) in 8×8 blocks to reserve room for data hiding. It employs count-encryption so repeated encryption of an image is not statically predictable, then allows the hidden data to be extracted and the original image reconstructed .

The present project transforms this algorithmic work into a realistic final-year-project platform. It preserves the paper's core reversible-hiding idea but adds a focused security improvement: authenticated recovery. The system will verify the authenticity and integrity of the marked encrypted artifact before extraction or recovery. It will also operationalise the method through patient-scoped access, a protected patient-key reference, secure server-side processing, and immutable audit records.

2. Motivation and Problem Statement

2.1 Motivation

Medical images often travel between healthcare professionals, systems, and storage services. Protecting confidentiality is necessary, but a useful medical-image workflow also needs a reliable way to attach limited protected information—such as a concise diagnostic note, structured metadata, an authorization marker, or an artifact reference—without permanently altering the underlying image. Exact recovery is important because even a minor untracked pixel modification can be unacceptable in a research or clinical-image handling context.

The research paper provides a promising algorithmic baseline for creating reserved room in encrypted grayscale medical images . However, a practical final-year project should answer two additional questions. First, can the artifact be recognised as authentic before a recovered image is trusted? Second, how can patient-specific access, workflow accountability, and safe key use be managed when several users interact with the same patient record?

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2.2 Problem statement

Existing encrypted-image data-hiding approaches can achieve confidentiality, payload embedding, and reversibility, yet encryption statistics alone do not prove that the received encrypted artifact or embedded payload has not been changed. Recent work has explicitly addressed this limitation by coupling reversible encrypted-image data hiding with ciphertext and plaintext authentication . In addition, an algorithmic research paper alone does not specify the access controls, auditability, data model, and patient-record lifecycle required for a credible multi-user system.

Therefore, the problem addressed by this project is:

How can a secure, role-governed platform implement reversible data hiding in encrypted 8-bit grayscale medical images, preserve exact recovery, detect tampered artifacts before trusted recovery, and manage patient-scoped access without exposing patient encryption keys in the user interface?

3. Objectives

3.1 Primary objective

To design and evaluate a role-governed RDHEMI platform that protects grayscale medical images, embeds a protected payload, verifies artifact integrity, and recovers the original image exactly after authorised extraction.

3.2 Specific objectives

- 1 Implement the paper's baseline RDHEMI workflow for supported 8-bit single-frame grayscale image matrices, including MED prediction, canonical/adjusted Huffman-tree coding, HC/LLC selection, count-encryption, data hiding, extraction, and recovery.
- 2 Create an authenticated extension using a protected manifest and compact integrity tag so tampered artifacts are rejected before they are represented as trusted.
- 3 Implement Trusted Authority, Doctor, and Patient roles with server-enforced patient-scoped access.
- 4 Create a duplicate-safe patient registry in which the Trusted Authority creates canonical profiles and Doctors add approved patients to their own care lists.

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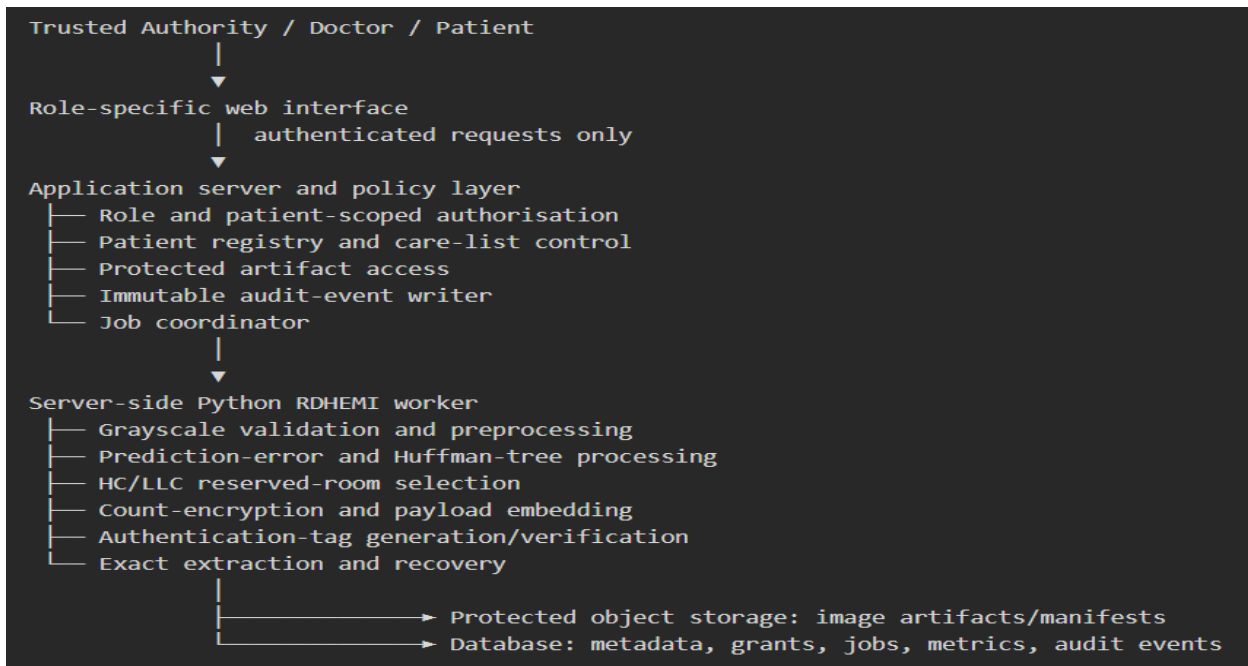
- 5 Store original, encrypted, and marked encrypted image artifacts through a protected server-side storage layer; do not expose patient keys or unrestricted storage URLs in the interface.
- 6 Provide a side-by-side original/recovered comparison viewer showing pixel-difference count and checksum verification.
- 7 Record immutable audit events for profile creation, access approvals, image operations, extraction, recovery, and integrity failures.
- 8 Evaluate capacity, exact reversibility, integrity detection, encryption behaviour, runtime, and role-based security using de-identified public grayscale datasets.

4. System Architecture / Proposed Methodology

4.1 Architectural overview

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The proposed platform follows a secure client–server structure. The web interface displays only permitted patient information, job status, image previews, metrics, and recovery results. It does not receive patient encryption keys. A server layer enforces authentication, role checks, patient-specific access grants, audit logging, and protected file delivery. A server-side Python RDHEMI worker processes the image matrix and produces algorithm artifacts. The database stores metadata, permissions, job state, metrics, and append-only audit records. Image artifacts and full manifests are stored in protected object storage and are delivered only after a server-side authorisation check.



4.2 User roles and workflow boundaries

Role	Core responsibility	Permitted high-level actions
Trusted Authority	Governance and identity management	Create Doctor and Patient profiles; validate new patient-registration requests; approve or deny patient-scoped Doctor access; review audit history.
Doctor	Authorised image protection and recovery workflow	Search the canonical patient directory; add approved patients to a care list; upload supported image; run encrypt/hide/extract/recover operations within granted scope.
Patient	Transparency and permitted viewing	View approved image/recovery history and relevant access activity through a restricted read-only view in the MVP. <i>Devendra</i>

The project separates a user profile from a clinical patient record. A patient has one canonical record with one stable protected key reference and longitudinal artifact history. Different Doctors can add the same approved patient to separate care lists, but they cannot create duplicate patient identities or independent patient keys.

4.3 Proposed RDHEMI processing workflow

The methodology will implement the supplied paper as the reference baseline and then apply the authenticated extension.

Stage	Baseline RDHEMI activity	Proposed authenticated extension
1. Input validation	Confirm supported single-frame 8-bit grayscale matrix	Reject unsupported colour, multi-frame, incompatible pixel-depth, or malformed inputs before processing.
2. Prediction	Generate prediction errors using MED	Preserve the chosen profile/version in a manifest.
3. Room reservation	Build/adjust canonical Huffman tree; compare HC and LLC per 8×8 block; retain shorter representation	Optional later experiment: compare fixed 8×8 against metadata-aware adaptive block sizing.

4. Key derivation	Apply the patient-associated key and count-encryption logic	Derive purpose-separated encryption, payload, and authentication subkeys using a job nonce and key version.
5. Encryption	Produce an encrypted image of the same dimensions	Include algorithm profile, dimensions, encryption count, and checksums in a protected canonical manifest.
6. Payload embedding	Encrypt and embed a short report/metadata payload in reserved room	Store a compact integrity tag that binds manifest and marked encrypted artifact.
7. Extraction/recovery	Extract payload and reconstruct original image for an authorised operation	Verify integrity tag first; stop safely and audit a failure if the artifact does not verify.
8. Verification	Compare source/recovered matrices	Display zero pixel mismatch and matching checksums only after successful exact verification.

4.4 Authenticated recovery mechanism

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The authenticated extension treats integrity as a separate security property from confidentiality. For each processing job, the system creates a canonical manifest containing the algorithm profile version, image dimensions, source checksum, patient-key version identifier, encryption count, job nonce, payload length/checksum, coding-metadata checksum, and timestamp. A compact integrity tag is computed with an authentication subkey over the manifest and marked encrypted artifact.

```

K_enc      = KDF(patientRootKey, jobNonce, "rdhemi-encryption-v1")
K_payload  = KDF(patientRootKey, jobNonce, "rdhemi-payload-v1")
K_auth     = KDF(patientRootKey, jobNonce, "rdhemi-authentication-v1")

integrityTag = HMAC(K_auth, canonicalManifest || markedEncryptedImageBytes)

```

The user interface never displays these key values. Before data extraction or recovery, the worker recalculates and verifies the tag. If verification fails, the system records an integrity failure and does not label any result as trusted recovery. This design follows the broader direction of recent reversible encrypted-image authentication methods while remaining a defined project extension to the supplied baseline .

4.5 Optional capacity-improvement experiment

The fixed 8×8 HTC-CE implementation is the mandatory baseline. If the baseline is stable, the project will evaluate one optional capacity extension: choose a block profile from 4×4, 8×8, or 16×16 based on the total cost of coding plus side information. The decision must account for every profile flag, map bit, and tree-description bit. The primary metric will be net payload bpp, not gross reserved room.

```
Net payload bits
= reserved-room bits
- tree/code description bits
- block/profile map bits
- header and payload framing bits
- authentication-tag bits

Net embedding rate = net payload bits ÷ (image width × image height)
```

5. Dataset Description

The project will use publicly available, de-identified grayscale medical-image data. No private patient image will be used without institutional approval and appropriate governance.

5.1 Primary benchmark dataset

The primary planned source is the NIH Chest X-ray dataset. Official documentation describes it as a collection of 100,0 de-identified chest X-ray images available in PNG form, with corresponding DICOM instances available through the documented public-data path . Its grayscale image content and availability in both PNG and DICOM representations make it suitable for an 8-bit input-profile benchmark.

For a final-year project, the evaluation will use a reproducible subset rather than the entire collection. The target is a stratified subset of 300 images selected by a fixed random seed and published image-ID list. The selection will include normal and abnormal examples where label availability permits, but the research evaluates reversible hiding/security properties—not disease detection performance.

5.2 Secondary generalisation dataset

Subject to time and access constraints, a smaller grayscale CT or MRI subset will be sampled from The Cancer Imaging Archive (TCIA). TCIA hosts de-identified medical-imaging collections accessible for public download . This secondary benchmark will test whether capacity and runtime patterns differ across chest X-ray, CT, and MRI image textures.

Dataset group	Proposed use	Format focus	Purpose
NIH Chest X-ray	Primary benchmark; reproducible 300-image subset	8-bit grayscale PNG; validated DICOM comparison where feasible	Baseline capacity, exact recovery, tamper tests, and platform demonstration
TCIA CT/MRI	Optional secondary set	Validated grayscale slices after documented normalisation	Modality/generalisation comparison

6. Dataset Preprocessing

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Preprocessing must be deterministic, documented, and reversible relative to the matrix used by RDHEMI. The project will preserve the original uploaded artifact separately and calculate a source checksum before processing.

Step	Procedure	Rationale
1. File validation	Accept only a supported PNG or single-frame grayscale DICOM input profile; reject colour, multi-frame, malformed, and unsupported pixel-depth inputs.	Keeps the MVP aligned with the 8-bit grayscale RDHEMI method.
2. De-identification check	Use only public de-identified dataset images; do not import private clinical images into the project.	Minimises privacy risk in academic experimentation.
3. Pixel-matrix extraction	Decode the image server-side and obtain a single 8-bit grayscale intensity matrix.	Provides the exact pixel-domain input for prediction, coding, encryption, and recovery.
4. Metadata separation	Keep DICOM/PNG source metadata outside the hidden image payload; retain only the approved manifest fields for processing.	Prevents accidental embedding of unnecessary sensitive information.

5. Dimension policy	Do not resize, crop, or apply lossy compression to the RDHEMI input matrix.	Makes exact recovery meaningful and avoids unreported image alteration.
6. Checksum creation	Compute a cryptographic checksum of the normalised input matrix and record it in the manifest.	Enables independent source/recovered verification.
7. Dataset split record	Store image IDs, source, preprocessing profile version, and random seed.	Supports reproducibility.

Exact-recovery definition: The project promises exact recovery of the documented, normalised 8-bit grayscale matrix supplied to the RDHEMI algorithm. The original source file is retained separately. The prototype does not claim arbitrary DICOM-file byte-for-byte reversibility across all transfer syntaxes, pixel depths, and metadata representations.

7. Pending Tasks / Work in Progress

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The project has completed conceptual research, product scope, workflow design, role model, high-level design, research-improvement planning, and visual architecture documentation. The following work remains.

Work package	Current status	Remaining work
Literature review and baseline interpretation	In progress	Finalise related-work comparison and supervisor-approved novelty statement.
Dataset acquisition and record	Pending	Download/access permitted public subset; produce dataset manifest and preprocessing log.
RDHEMI baseline implementation	Pending	Implement deterministic 8-bit grayscale MED, HTC, HC/LLC selection, count-encryption, hiding, extraction, and exact recovery.
Golden test vectors	Pending	Create small known input/output tests for every reversible stage.

Role-governed web platform	Pending	Implement role checks, canonical patient registry, care lists, approval workflow, protected artifact delivery, and audit logging.
User interface	Pending	Build Authority, Doctor, Patient, Protect Image, and Recovery Comparison screens.
Capacity extension	Conditional	Add adaptive block-profile experiment after fixed 8×8 baseline is stable.
Evaluation and analysis	Pending	Run baseline/extension comparison, tamper tests, runtime profiling, and results interpretation.
Dissertation materials	Pending	Write methodology, results, limitations, references, demo script, and viva preparation notes.

8. Timeline / Work Plan

The following twelve-week plan is designed for an academic final-year-project cycle. Weeks may be shifted to align with the department calendar without changing the dependency order.

Week	Work plan	Main deliverable
1	Confirm synopsis, scope, success criteria, and source-paper baseline; finalise dataset-access plan.	Approved synopsis and requirements baseline
2	Acquire public grayscale data; generate dataset manifest; implement validation and preprocessing pipeline.	Reproducible dataset subset and preprocessing report
3	Implement MED prediction and prediction-error matrix generation; create unit tests.	Tested prediction module
4	Implement canonical/adjusted Huffman-tree handling, HC/LLC coding, and fixed 8×8 reserved-room calculation.	Tested HTC module
5	Implement count-encryption, payload embedding, extraction, and exact-recovery path.	Baseline RDHEMI end-to-end prototype

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6	Generate golden test vectors; verify zero pixel mismatch; measure initial gross/net bpp and runtime.	Baseline evaluation report
7	Implement authenticated manifest, domain-separated keys, integrity tag, and tamper-failure handling.	Authenticated RDHEMI prototype
8	Implement database schema, roles, canonical patient records, care lists, Doctor access requests, and immutable audit events.	Secure backend workflow
9	Implement protected artifact storage and Python-job coordination; integrate security-metrics capture.	Server-side processing workflow
10	Build role-based screens and the original/recovered side-by-side viewer.	Usable end-to-end web demonstration
11	Run full benchmarks, attack tests, role-denial tests, and performance analysis; optionally evaluate adaptive block sizing.	Results tables, charts, and findings
12	Complete dissertation chapters, user guide, demonstration script, limitations, and viva preparation.	Final report and demonstration-ready system

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9. Expected Outcomes

The project is expected to deliver the following research and software outcomes.

Category	Expected outcome
Algorithm baseline	A reproducible implementation of the supplied HTC-CE RDHEMI method for the supported 8-bit grayscale input profile.
Security extension	An authenticated-recovery mechanism that detects changes to the marked encrypted artifact or its protected metadata before trusted extraction/recovery.
Exactness evidence	Zero pixel mismatch and matching matrix checksums between supported input and recovered output in successful test cases.
Capacity evidence	Gross room, metadata overhead, authentication overhead, and net payload bpp reported transparently.

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Performance evidence	Encryption, hiding, extraction, and recovery runtime and memory measurements on the stated benchmark.
Platform prototype	A role-governed final-year-project web platform with Authority, Doctor, and Patient workflows, patient-scoped access, protected artifacts, job history, security metrics, and audit records.
Clinical demonstration	A side-by-side recovery viewer that visibly demonstrates original/recovered agreement and recorded integrity status.
Research documentation	Dataset/preprocessing manifest, algorithm profile, test cases, attack-test results, methodology, limitations, and reproducibility notes.

9.1 Measurable success criteria

Criterion	Target
Exact recovery on supported inputs	Pixel mismatch count = 0 and recovered matrix checksum equals source matrix checksum
Tamper handling	Deliberate modification of protected artifact, manifest, or payload causes verification failure before any result is labelled trusted
Key privacy	Patient root key is never returned by a client-facing API, displayed in the interface, or recorded in normal application logs
Authorisation	A Doctor without active patient-scoped permission cannot start sensitive processing or retrieve protected artifacts
Auditability	Every sensitive event creates an append-only event record with actor, patient scope, action, result, and timestamp
Capacity reporting	Net bpp is reported after all coding, map, header, and integrity overhead is included

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10. Limitations and Scope Boundaries

The first version deliberately focuses on deterministic, 8-bit, single-frame grayscale image matrices. It will not initially support colour images, multi-frame studies, all 12/16-bit DICOM modalities, compressed DICOM transfer syntaxes, enterprise PACS integration, hardware security modules, patient-managed consent, automatic ROI segmentation, or regulatory certification. These are important future directions, but including them in the first version would reduce the likelihood of a rigorous, testable, and complete final-year project.

The project will report encryption statistics such as entropy, correlation, chi-square, NPCR, UACI, and PSNR where applicable. These metrics describe empirical behaviour; they do not alone constitute a complete cryptographic proof. The design therefore combines statistical reporting with explicit integrity verification, access-control tests, key-handling restrictions, and tamper scenarios.

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11. References

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12. Acknowledgement

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I acknowledge the authors of the baseline RDHEMI research paper and the organisations that make de-identified public medical-image datasets and standards documentation available for research and education. Finally, I am grateful to my classmates, friends, and family for their encouragement and support.

Appendix A. Suggested Final Project Title Options

Option	Title
1	Secure Reversible Data Hiding in Encrypted Grayscale Medical Images: An Authenticated, Role-Governed RDHEMI Platform
2	Authenticated RDHEMI for Secure Grayscale Medical Image Sharing and Exact Recovery
3	AT-HTC-CE RDHEMI: Authenticated Huffman Tree Coding and Count-Encryption for Reversible Medical Image Data Hiding

Appendix B. Demonstration Story

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- 1.The Trusted Authority creates Doctor and Patient profiles.
 - 2.The Doctor signs in, finds the canonical patient record, and receives patient-scoped access.
 - 3.The Doctor uploads a supported grayscale image.
 - 4.The server-side worker protects the image using RDHEMI and stores the artifact privately.
 - 5.The Doctor attaches a protected short clinical note when capacity permits.
 - 6.An authorised user starts extraction/recovery.
 - 7.The worker verifies integrity; if verification succeeds, it extracts the note and recovers the original matrix.
 - 8.The comparison viewer shows the original and recovered images, zero pixel difference, matching checksums, and the corresponding immutable audit trail.
- If the marked encrypted image is deliberately changed, the integrity check fails, recovery is not labelled trusted, and the event is recorded in the audit log.